

Almost Here: Self-Learning, Self-Healing Computers

Processors that mimic the human brain could be the next big disruption in computing

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How long can the industry rely on Moore's Law? Today's computers are rather too good at calculating stuff and can achieve anything that can be reduced to a numerical problem. But complex problems, which need a good amount of reasoning or those that need so-called 'intuition,' require too much programming and hence too much processing and power too. Just stuffing more transistors into smaller chips will take us nowhere. So, what next?

Seeking to move to the next new frontier of computing, research teams across the world are trying to move away from traditional chip designing methods and radically redesign memory, computation and communication circuitry based on how the neurons and synapses of the brain work. This will be a big leap in artificial intelligence, eventually resulting in self-learning computers that will be able to understand and adapt themselves to changes, complete tasks without routine programming and work around failures too. Such self-learning computers are commonly dubbed as 'neuromorphic' as they mimic the human brain. Here, we look at some of the significant strides in this direction.

Mimicking the mammalian brain in function, size and power consumption

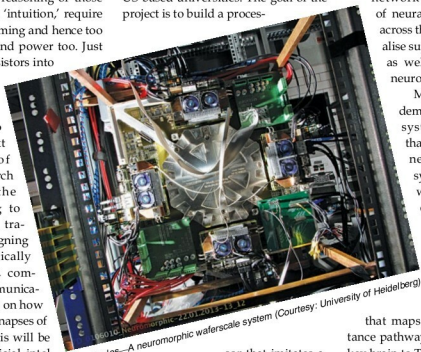
One of the largest and oldest projects in this direction is the DARPA sponsored Systems of Neuromorphic Adaptive Plastic Scalable Electronics (SyNAPSE), which is contracted mainly to IBM and HRL, along with some US-based universities. The goal of the project is to build a proces-

Both the cores were fabricated in 45nm silicon on insulator (SOI) complementary metal oxide semiconductor (CMOS) and contained 256 neurons. One core had 262,144 programmable synapses while the other had 65,536 learning synapses. Then came the Brain Wall—a visualisation tool that allows researchers to view neuron activation states in a large-scale neural network and observe patterns of neural activity as they move across the network. It helps visualise supercomputer simulations as well as activities within a neuromorphic core.

Meanwhile, in 2012, IBM demonstrated a computing system called TrueNorth that simulated 530 billion neurons and 100 trillion synapses, running on the world's second-fastest operating supercomputer. TrueNorth was supported by Compass—a multi-threaded, massively parallel functional simulator and a parallel compiler

that maps a network of long-distance pathways in the macaque monkey brain to TrueNorth.

Last year, they had more updates. IBM revealed that the chips are radically different from the current Von Neumann architecture based ones. The new model works with multiple low-power processor cores working in parallel. Each neuromorphic core has its own memory (synapses), a processor (neuron) and communication conduit (axon). By operating these suitably, one can achieve recognition



BrainScaleS—A neuromorphic waferscale system (Courtesy: University of Heidelberg)

sor that imitates a mammal's brain in function, size and power consumption. Specifically, "It should recreate 10 billion neurons, 100 trillion synapses, consume one kilowatt and occupy less than two litres of space." Since it started in 2008, the project has seen some interesting results.

The first breakthrough came in 2011, when IBM revealed two working prototypes of neuromorphic chips.